

## Demonstration of Proposed Features

|                      |                                                      |
|----------------------|------------------------------------------------------|
| <b>Date</b>          | 14 July 2026                                         |
| <b>Team ID</b>       | XXXXXX                                               |
| <b>Project Name</b>  | OptiCrop – AI Intelligent Crop Recommendation System |
| <b>Maximum Marks</b> | 1 Mark                                               |

### Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

| S.No | Feature Name                | Description                                                                            | Status (Implemented / Partial / Pending) | Demonstrated (Yes / No) | Remarks                      |
|------|-----------------------------|----------------------------------------------------------------------------------------|------------------------------------------|-------------------------|------------------------------|
| 1    | Crop Recommendation         | Predicts the most suitable crop using soil nutrient and environmental parameters.      | Implemented                              | Yes                     | Core functionality completed |
| 2    | Machine Learning Prediction | Random Forest model trained to generate accurate crop recommendations.                 | Implemented                              | Yes                     | Successfully tested          |
| 3    | User-Friendly Web Interface | Responsive web interface for entering agricultural parameters and viewing predictions. | Implemented                              | Yes                     | Easy to use                  |
| 4    | Data Visualization          | Displays dataset insights and feature relationships using charts.                      | Implemented                              | Yes                     | Improves analysis            |
| 5    | Prediction Report           | Shows predicted crop with confidence and recommendation summary.                       | Implemented                              | Yes                     | Working successfully         |
| 6    | Model Evaluation            | Performance evaluated using Accuracy, Precision, Recall and F1-Score.                  | Implemented                              | Yes                     | Verified during testing      |

### Feature Implementation Summary:

|                                   |   |
|-----------------------------------|---|
| <b>Total Features Proposed</b>    | 6 |
| <b>Total Features Implemented</b> | 6 |

|                                 |      |
|---------------------------------|------|
| Total Features Demonstrated     | 6    |
| Overall Implementation Rate (%) | 100% |